

SAMIKSHA BARNWAL

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Education

Lovely Professional University, Punjab	2022 – 2025
Bachelor of Computer Applications (BCA)	CGPA: 7.9

Technical Skills

Languages: Python, JavaScript
AI / ML: Machine Learning, NLP, Generative AI, Prompt Engineering, Model Evaluation
LLM Systems: LLM APIs, Embeddings, RAG, Agentic Systems, Planner–Executor–Verifier, LLM Evaluation Pipelines
Search & Retrieval: Semantic Search, Dense Retrieval, Vector Search, FAISS, Pinecone, Weaviate, Chroma
Backend & APIs: FastAPI, Flask, Node.js, REST APIs, WebSockets
Libraries: Pandas, NumPy, Scikit-learn, VADER
Cloud / DevOps: AWS, Docker, CI/CD, GitHub Actions
Collaboration: Git, GitHub, Agile, Startup-paced execution

Projects

AI Operations Assistant | Multi-Agent GenAI, FastAPI, Retrieval

- Engineered a planner–executor–verifier multi-agent system automating **20+ workflows**, reducing manual resolution time by **30+ minutes per workflow**.
- Built RAG pipelines with embeddings and vector search, improving knowledge retrieval speed by **40%** for reasoning-heavy operational tasks.
- Designed verifier loops and LLM evaluation checkpoints to improve output consistency and agent reliability.

YouTube Comment Sentiment & Toxicity Intelligence | NLP, Evaluation

- Processed **5,000+ multilingual comments** with **85% accuracy** using sentiment and toxicity pipelines.
- Reduced toxicity false positives by **15%** through threshold tuning, normalization, and ambiguity-focused error analysis.

Rose Plant Disease Intelligence System | CNN Transfer Learning, CV, Data Analysis

- Built a disease detection pipeline over **5,000+ rose leaf images**, combining transfer learning with deep exploratory data analysis and preprocessing audits.
- Benchmarked MobileNetV2 vs EfficientNetB3, achieving **83.2% accuracy**, **0.83 F1-score**, and **0.82 recall**; outperformed baseline by **7.6 points**.
- Performed class distribution analysis, augmentation experiments, confusion-matrix debugging, and failure-case evaluation to improve generalization.
- Extended the model into a real-time diagnosis web app with confidence scoring and treatment recommendations.

ML-Based Threat Detection System | Splunk MLTK, Anomaly Detection

- Built anomaly detection over **1GB+ security logs**, detecting abnormal VPN and USB behavior.
- Reduced training time by **20%** via optimized preprocessing and feature filtering.

Achievements

- Winner – IBM Datathon 2024
- 3rd Place – IBM Cloud Competition
- Certifications: IBM Data Analysis, Coursera Machine Learning, SQL for Data Science